

ELECENG 2E14: Electronic Devices and Circuits 1

Design Project 4

Anushka Chauhan - Chauha30 - 400598874

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Circuit schematic

Boolean expression for XOR Gate

$$Y = A \oplus B$$

$$Y = \bar{A}B + \bar{B}A$$

→ apply De Morgan's theorem

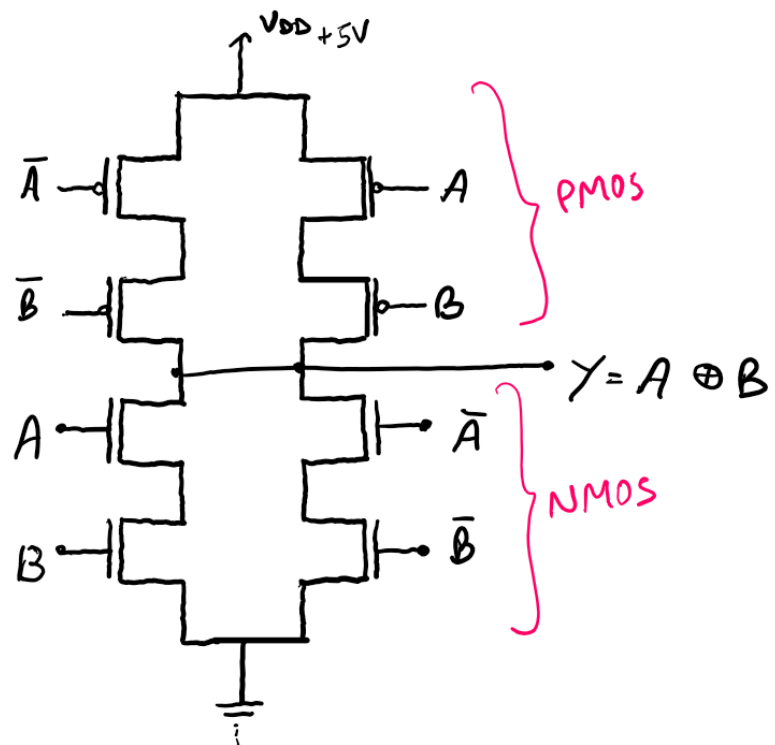
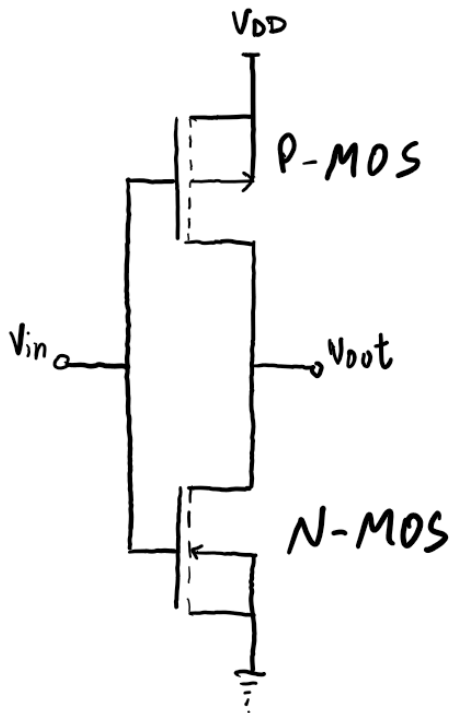
$$Y = \overline{\bar{A}B + \bar{B}A}$$

$$Y = \overline{(\bar{A}B) \cdot (\bar{B}A)}$$

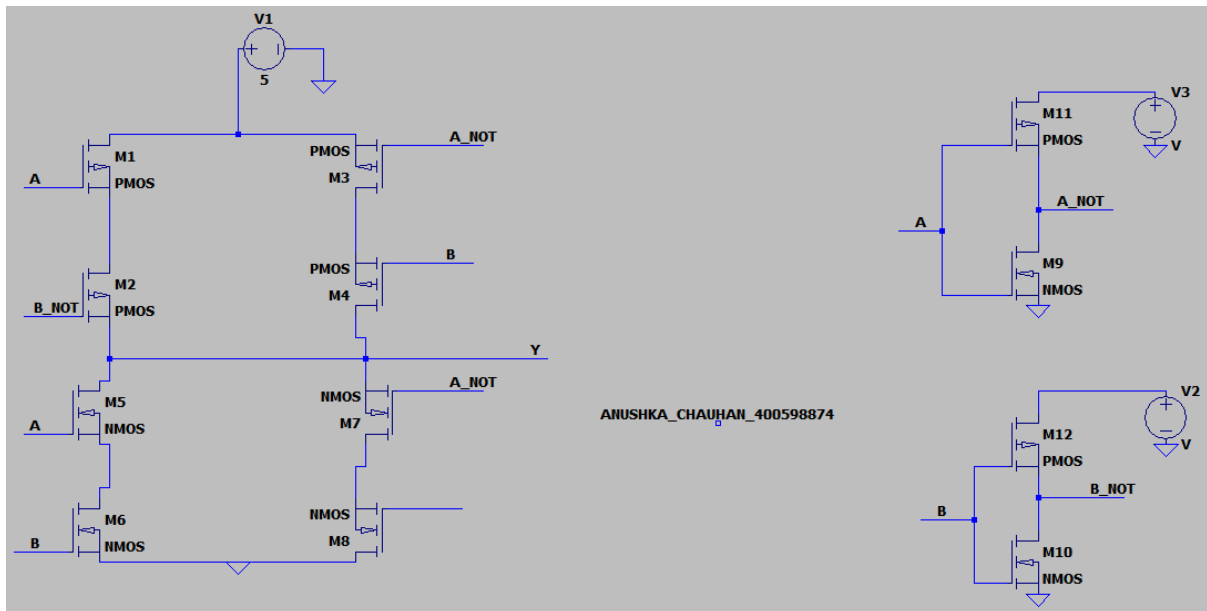
$$Y = \overline{(A + \bar{B}) \cdot (\bar{A} + B)}$$

$$Y = \overline{AB + \bar{A}\bar{B}}$$

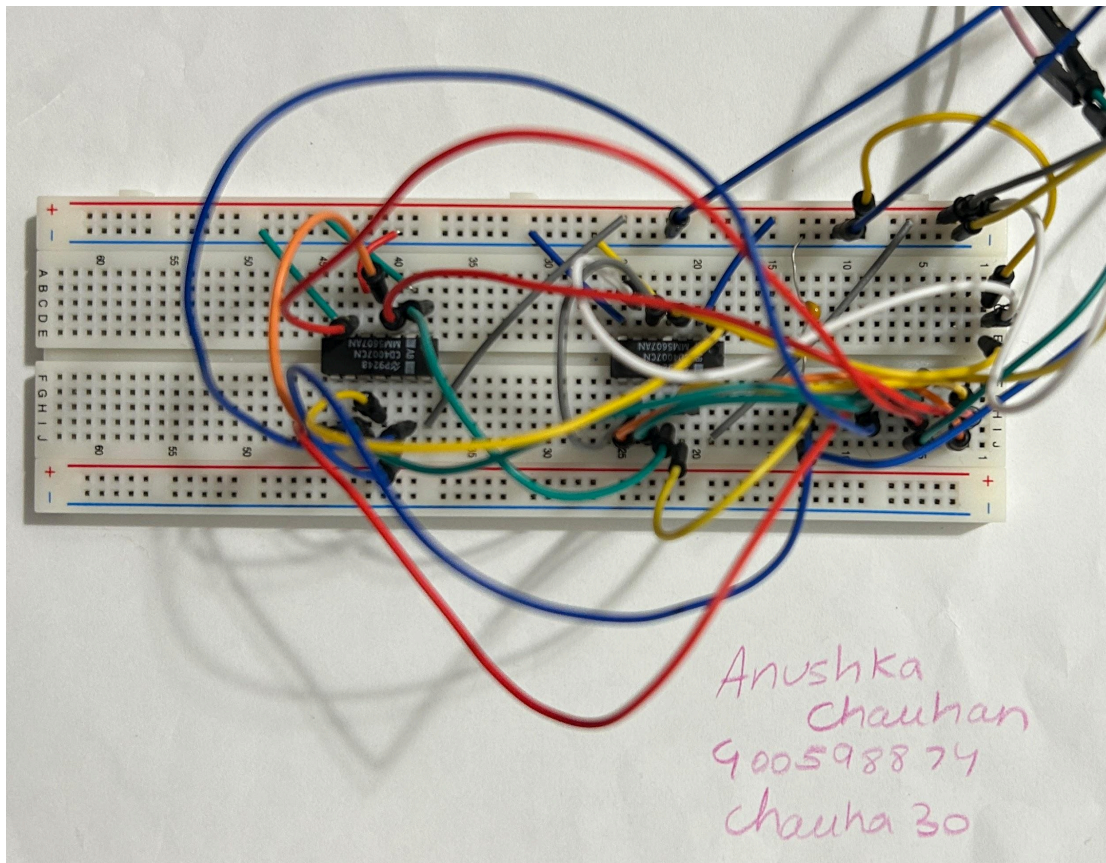
CMOS Inverter CMOS XOR GATE



LTspice Schematic:



Physical circuit picture:



Ideal sizing

When designing a CMOS gate, it's important for the output to switch from low to high as quickly as it does from high to low. However, PMOS and NMOS transistors have different speeds because they operate using different charge carriers. NMOS transistors use electrons, which move faster, while PMOS transistors use "holes," or the absence of slower electrons. In fact, electrons are about 2.5 times quicker than holes.

If both transistors were the same size, the PMOS would be slower, leading to an unbalanced circuit where one transition takes longer than the other. To fix this, we make the PMOS transistor larger to offset its slower speed. Specifically, we use a size ratio of 2.5:1, meaning the PMOS should be 2.5 times wider than the NMOS. In practical terms, this could mean a PMOS width-to-length ratio of 5/1 and an NMOS ratio of 2/1 based on a standard inverter size.

This increase in the PMOS size helps balance its performance, so both transistors switch on and off at similar rates, resulting in a more efficient circuit. For an XOR gate, the worst-case scenario can involve two transistors in series, so each one needs to be doubled in size (2n for NMOS and 2p for PMOS). This ensures that when their resistances combine, they still match the resistance of the reference inverter.

Implementation of ideal sizing in the hardware

In simulation tools like LTspice, it's possible to adjust transistor sizes to match an ideal design. However, when working with actual hardware, such as the CD4007B chip, a significant limitation arises. The transistors within that chip come with fixed dimensions that cannot be altered.

This limitation is crucial because the ideal PMOS to NMOS transistor ratio is typically around 2.5:1, but the manufacturers of the CD4007B may not have designed it with this ratio in mind. If the PMOS transistors are smaller than needed, increased resistance will slow down the transition from low to high output. This may result in an output that rises more slowly than it falls, which is not desirable. Conversely, if the PMOS transistors are oversized, the transition from low to high can be quicker than from high to low, leading to a different kind of imbalance in performance.

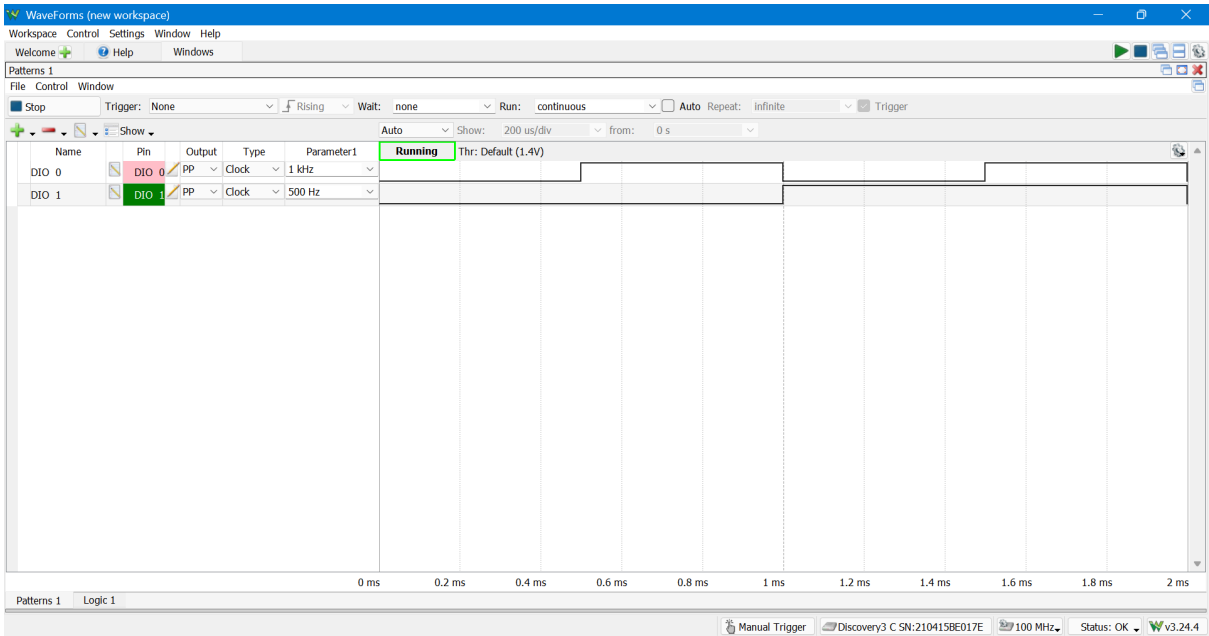
In both scenarios, the symmetry intended in the design is lost. This discrepancy impacts how quickly and efficiently the circuit operates. Measured timing for these transitions often doesn't align with expectations, and overall propagation delays can exceed the theoretical minimum. This highlights the challenges posed by using standard components, which can significantly influence circuit performance. While simulations offer a valuable preview, the practical realities of working with physical components can reveal limitations that affect overall design goals.

Functional testing

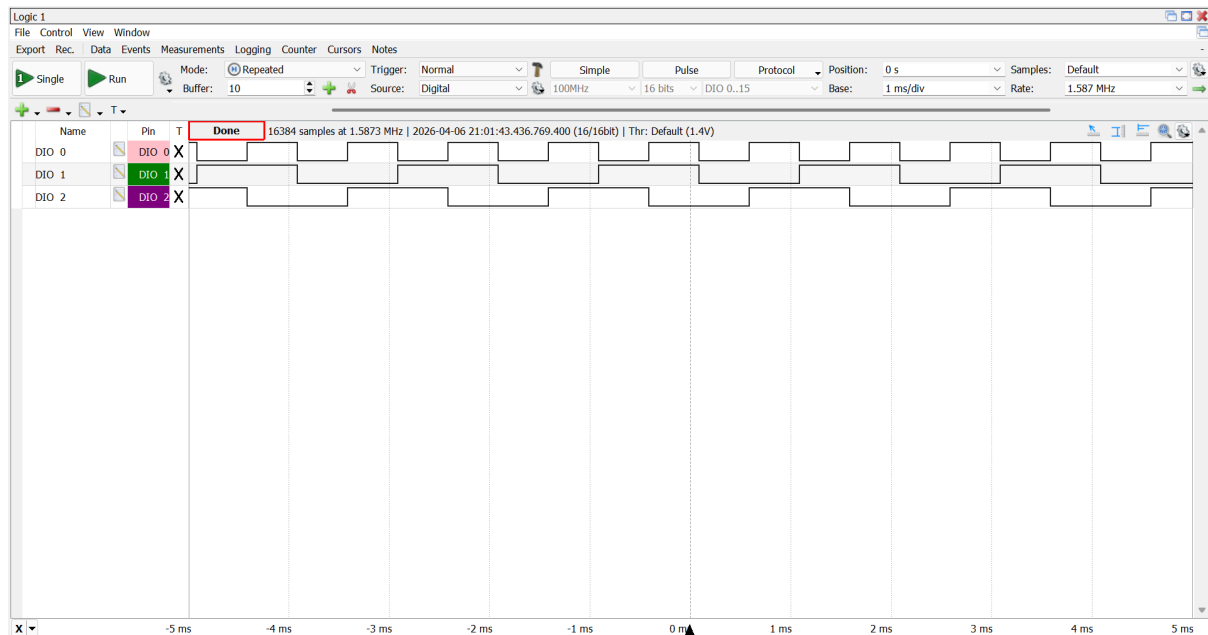
XOR Truth Table

A (Input 1)	B (Input 2)	$X = A'B + AB'$
0	0	0
0	1	1
1	0	1
1	1	0

The input through the pattern window into A (DIO 0) and B (DIO 1)



Results from logic analyzer (DIO0 = A, DIO1 = B, DIO2 = Y(output))



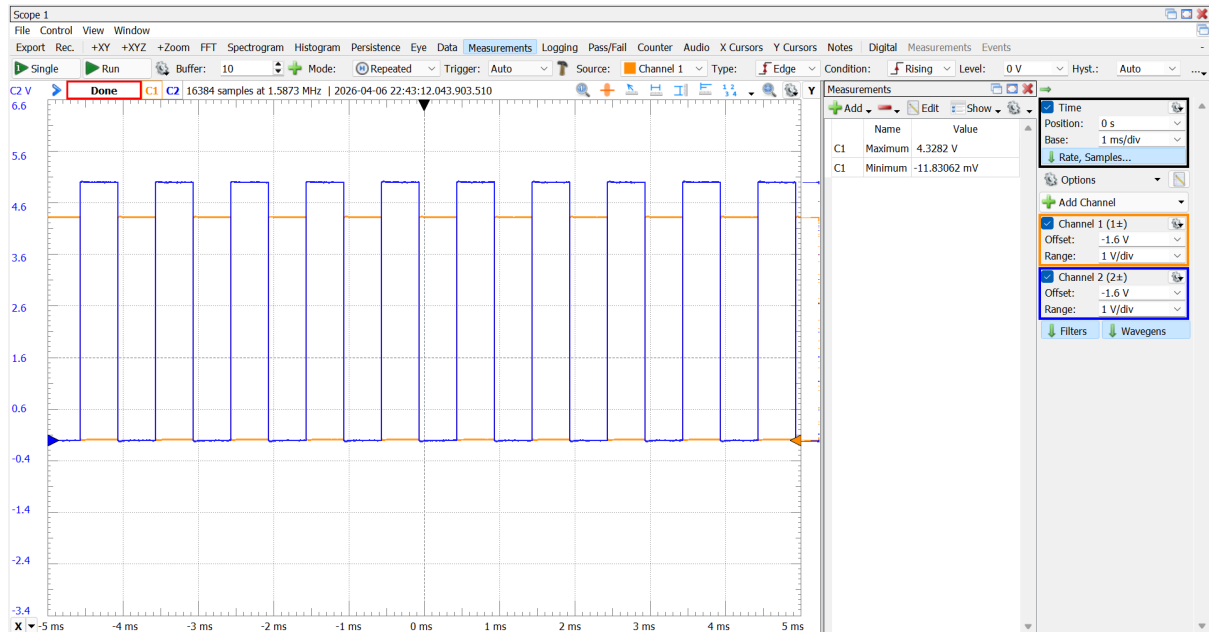
In this experiment, we used the Analog Discovery 2 (AD2) to check if our circuit operates as expected according to the XOR (exclusive OR) logic function. To do this, we set up a Pattern Generator which provided the necessary input signals. We configured two Digital Input/Output (DIO) channels: DIO 0 (Input A) was set to produce a 1 kHz square wave, while DIO 1 (Input B) produced a 500 Hz square wave. This specific frequency setup allowed the circuit to go through all possible combinations of the binary inputs (00, 01, 10, and 11) every two milliseconds. The output from our 12-transistor XOR circuit was connected to DIO 2 on the Logic Analyzer, allowing us to observe the output waveforms in real-time. Looking at the captured waveform, we can see that the circuit effectively demonstrates the XOR function. The output at DIO 2 shows a logic-high state (+5V) only when the two inputs are different (one high, one low). In contrast, when both inputs are either low or high at the same time, the output stays low (0V).

Static level testing

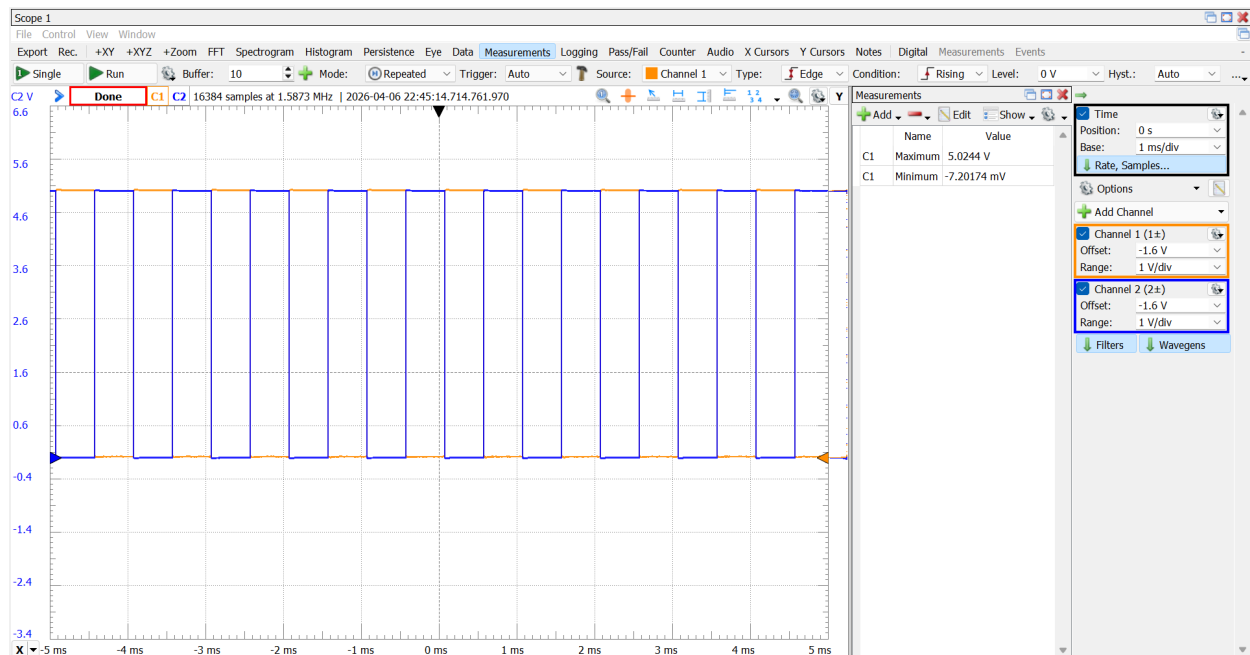
For this testing the wavegen is set at frequency = 1KHz, amplitude = 2.5V and offset = 2.5V
V+ supply is set to +5V

Channel 2 in input, where wavegen is connected, and Channel 1 is the output

The output when A is set to +5, and B is set to Wavegen 1



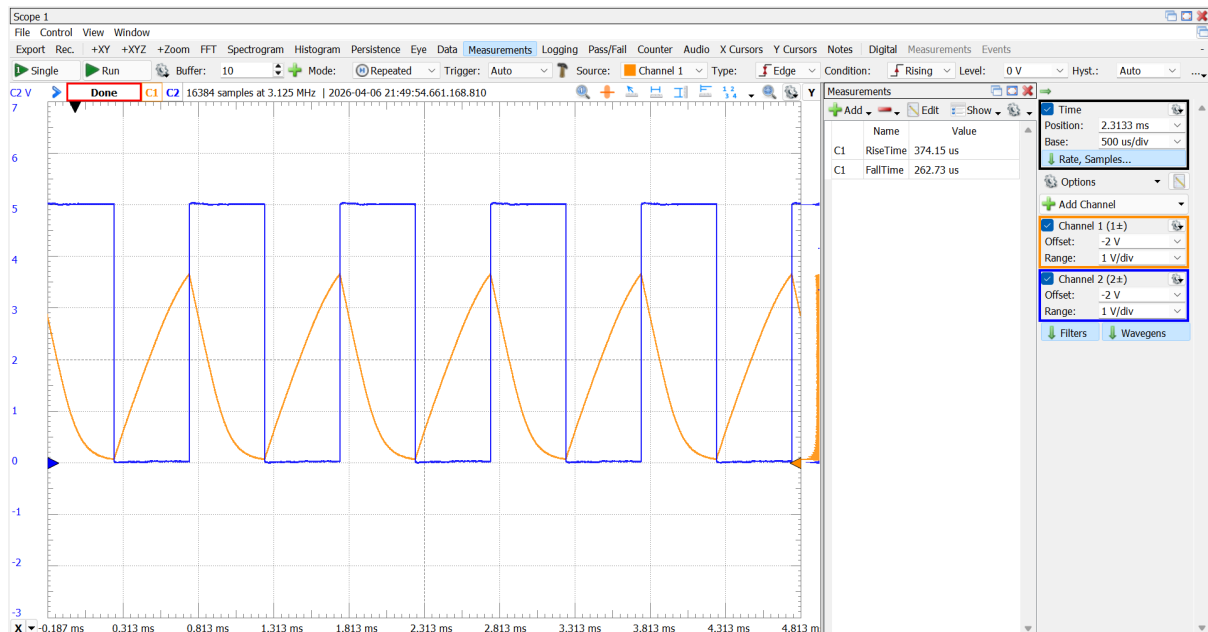
The output when A is set to wavegen and B is set to +5



The XOR gate shows consistent voltage levels for its output. In the first test, one input (A) was connected to a steady +5V, while the other(B) was controlled by a square wave. The results showed the output reaching a maximum voltage (VH) of about 4.32 V and a minimum (VL) of -11.83 mV. This indicates that the gate effectively gets the output close to the power supply levels. When the inputs were switched, moving the +5V to the other input (B), the output values were very similar, with VH at around 5.02 V and VL at -7.20 mV. The slight increase in the high voltage in this case suggests the gate is performing better as the power supply settles. The consistent shape of the output waveform and voltage levels after this switch shows that the internal components of the gate are evenly matched. This means the gate can effectively handle loads and maintain reliable performance no matter which inputs are used.

Timing

After adding the capacitor at output Y, when A is set to +5V and B is set to wavegen, the following graph is achieved.



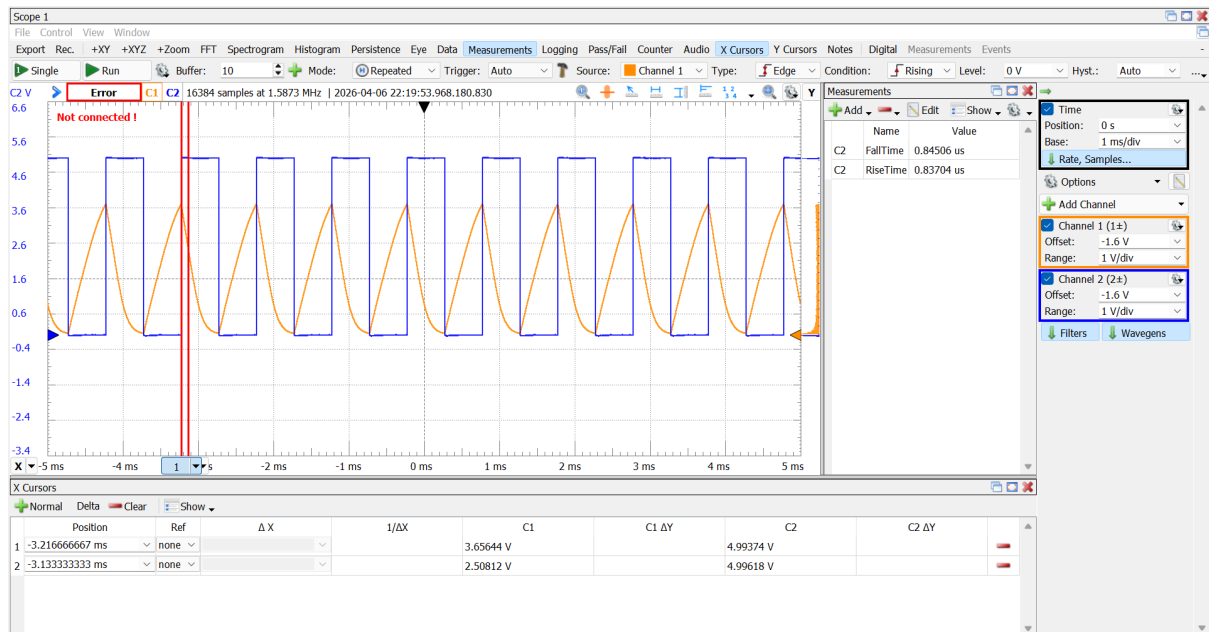
1. Rise Time measured from the graph is $374.15\mu\text{s}$, and fall time is measured as $262.73\mu\text{s}$

The rise time was measured at $374.15\mu\text{s}$, while the fall time was measured at $262.73\mu\text{s}$. The discrepancy between these two values is primarily due to the difference in carrier mobility between the PMOS (pull-up) and NMOS (pull-down) transistors. Since NMOS transistors typically have higher electron mobility, they can discharge the 100 nF capacitor faster than the PMOS transistors can charge it, resulting in a quicker fall time compared to the rise time.

2. Propagation delay calculation

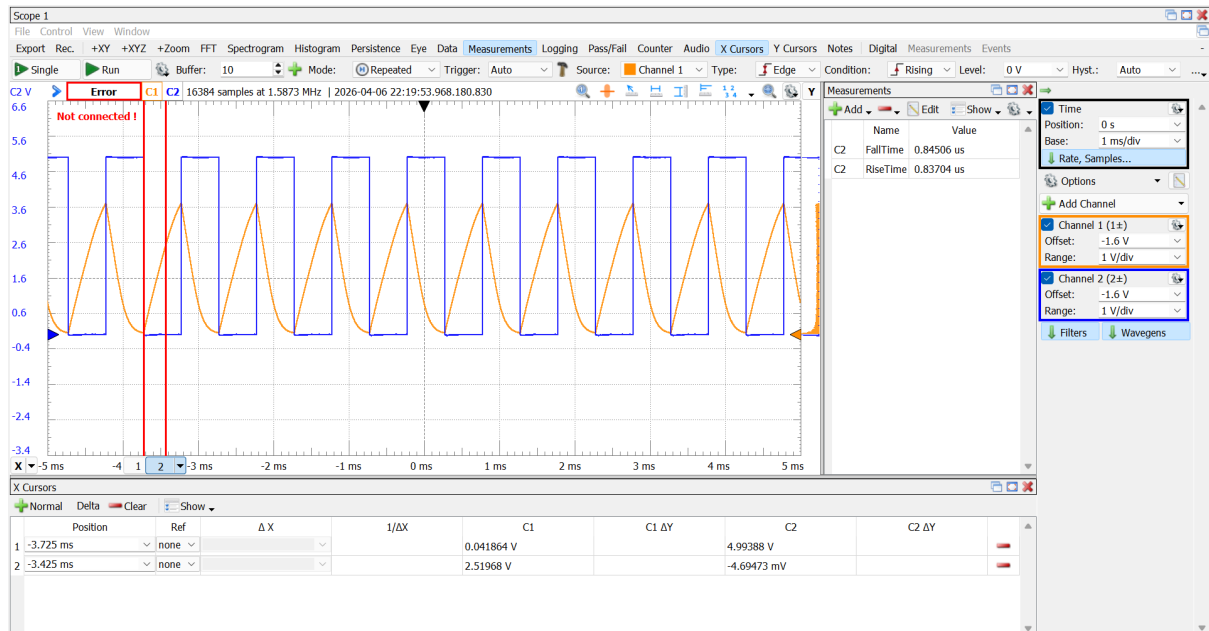
Cursor placement for High- to -Low Propagation Delay

Using cursor placement at the 50% points of the rising input and falling output, the delay was calculated as:



Cursor Placement for Low- to - High propagation delay

Using cursor placement at the 50% points of the falling input and rising output, the delay was calculated as:



The average propagation delay is $\tau_p = (\tau_{PHL} + \tau_{PLH})/2 = 191.65\mu\text{s}$

References

[1] A. S. Sedra, K. C. Smith, T. C. Carusone, and V. Gaudet, *Microelectronic Circuits*, 8th ed. New York, NY: Oxford University Press, 2019.

[2] "CD4007C Dual Complementary Pair Plus Inverter," Fairchild Semiconductor,
<https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/6380/1N4007.pdf>

[3] "Analog Discovery 2 - Digilent Reference," digilent.com. Accessed: Jan. 31, 2023.
[Online]. Available:
<https://digilent.com/reference/test-and-measurement/analog-discovery-2/start?redirect=1>